

$$e^{-\tau(H-E_*)^2} \quad = \quad \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-s^2/(4\tau)} e^{-is(H-E_*)}\, ds$$

Narrower energy window $\longleftrightarrow$ longer evolution times, $|s| \ \sim \ \sqrt{\tau}$